

Name: Sachin
Roll. No.: 251256
Dept: B.sc (H.) CS
Artificial Intelligence (AI) ASSIGNMENT: 2

6. Write a program in PROLOG to implement factorial (N, F) where F represents the factorial of a number N.

Practicals > P6-Factorial.pl

```

1 factorial(0, 1).
2
3 factorial(N, F) :-
4     N > 0,
5     N1 is N - 1,
6     factorial(N1, F1),
7     F is N * F1.
```

```

1 ?- ['P6-Factorial.pl'].
true.

2 ?- factorial(7,F).
F = 5040 .

3 ?- trace.
true.

[trace] 3 ?- factorial(7,F).
Call: (12) factorial(7, _19952) ? creep
Call: (13) 7>0 ? creep
Exit: (13) 7>0 ? creep
Call: (13) _23340 is 7+ -1 ? creep
Exit: (13) 6 is 7+ -1 ? creep
Call: (13) factorial(6, _25114) ? creep
Call: (14) 6>0 ? creep
Exit: (14) 6>0 ? creep
Call: (14) _27780 is 6+ -1 ? creep
Exit: (14) 5 is 6+ -1 ? creep
Call: (14) factorial(5, _29554) ? creep
Call: (15) 5>0 ? creep
Exit: (15) 5>0 ? creep
Call: (15) _32220 is 5+ -1 ? creep
Exit: (15) 4 is 5+ -1 ? creep
Call: (15) factorial(4, _33994) ? creep
Call: (16) 4>0 ? creep
Exit: (16) 4>0 ? creep
Call: (16) _36660 is 4+ -1 ? creep
Exit: (16) 3 is 4+ -1 ? creep
Call: (16) factorial(3, _38434) ? creep
Call: (17) 3>0 ? creep
Exit: (17) 3>0 ? creep
Call: (17) _41100 is 3+ -1 ? creep
```

```

PS D:\AI-Practicals-251256\Practicals> swipl
Call: (17) _41100 is 3+ -1 ? creep
Exit: (17) 2 is 3+ -1 ? creep
Call: (17) factorial(2, _42874) ? creep
Call: (18) 2>0 ? creep
Exit: (18) 2>0 ? creep
Call: (18) _45540 is 2+ -1 ? creep
Exit: (18) 1 is 2+ -1 ? creep
Call: (18) factorial(1, _47314) ? creep
Call: (19) 1>0 ? creep
Exit: (19) 1>0 ? creep
Call: (19) _49980 is 1+ -1 ? creep
Exit: (19) 0 is 1+ -1 ? creep
Call: (19) factorial(0, _51754) ? creep
Exit: (19) factorial(0, 1) ? creep
Call: (19) _47314 is 1*1 ? creep
Exit: (19) 1 is 1*1 ? creep
Exit: (18) factorial(1, 1) ? creep
Call: (18) _42874 is 2*1 ? creep
Exit: (18) 2 is 2*1 ? creep
Exit: (17) factorial(2, 2) ? creep
Call: (17) _38434 is 3*2 ? creep
Exit: (17) 6 is 3*2 ? creep
Exit: (16) factorial(3, 6) ? creep
Call: (16) _33994 is 4*6 ? creep
Exit: (16) 24 is 4*6 ? creep
Exit: (15) factorial(4, 24) ? creep
Call: (15) _126 is 5*24 ? creep
Exit: (15) 120 is 5*24 ? creep
Exit: (14) factorial(5, 120) ? creep
Call: (14) _122 is 6*120 ? creep
Exit: (14) 720 is 6*120 ? creep
Exit: (13) factorial(6, 720) ? creep
Call: (13) _62 is 7*720 ? creep
Exit: (13) 5040 is 7*720 ? creep
Exit: (12) factorial(7, 5040) ? creep
F = 5040 .
```

7. Write a program in PROLOG to implement generate_fib(N,T) where T represents the Nth term of the Fibonacci series.

Practicals > P7-Generate-fib.pl

```
1 generate_fib(0, 0).
2 generate_fib(1, 1).
3 generate_fib(N, T) :-
4     N > 1,
5     N1 is N - 1,
6     N2 is N - 2,
7     generate_fib(N1, T1),
8     generate_fib(N2, T2),
9     T is T1 + T2.
```

For online help and background, visit <https://www.swi-prolog.org>
For built-in help, use ?- help(Topic). or ?- apropos(Word).

```
1 ?- ['P7-Generate-fib.pl'].
true.
```

```
2 ?- generate_fib(8,T).
T = 21 .
```

```
3 ?- generate_fib(9,T).
T = 34 .
```

```
4 ?- 
```

8. Write a PROLOG program to implement power (Num, Pow, Ans) : where Num is raised to the power Pow to get Ans.

Practicals > P8-Power.pl

```
1 power(Num, Pow, Ans) :-  
2   Ans is Num ** Pow.
```

For online help and background, visit <https://www.swi-prolog.org>
For built-in help, use ?- help(Topic). or ?- apropos(Word).

```
1 ?- ['P8-Power.pl'].  
true.
```

```
2 ?- power(8,7,Ans).  
Ans = 2097152.
```

```
3 ?- trace.  
true.
```

```
[trace] 3 ?- power(8,7,Ans).  
  Call: (12) power(8, 7, _18936) ? creep  
  Call: (13) _18936 is 8**7 ? creep  
  Exit: (13) 2097152 is 8**7 ? creep  
  Exit: (12) power(8, 7, 2097152) ? creep  
Ans = 2097152.
```

THANK YOU